

Haodong Liang

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Education

University of California, Davis, USA. Ph.D. Electrical and Computer Engineering	Sep 2023 – Present
• Advisors: Lifeng Lai, Krishnakumar Balasubramanian • GPA: 4.0/4.0	
University of Michigan, Ann Arbor, USA. MSc. Statistics	Sep 2021 – May 2023
• GPA: 3.95/4.0	
Xiamen University, China. BSc. Mathematical Statistics	Sep 2016 – Jun 2020
• GPA: 3.64/4.0. Major GPA: 3.8/4.0.	

Research Interests

My research focuses on the theoretical and mechanistic foundations of in-context learning in Transformer models. I study how Transformers implement machine learning and decision-making algorithms in a provable view. My broader interests include continual learning, knowledge retention during fine-tuning, and differential privacy.

Research Projects

Provable In-Context Learning Capabilities of Transformer Models	Jan 2024 - Present UC Davis
• Studied the theoretical and mechanistic foundations of how a pretrained Transformer model learns and executes algorithms in context, supported by empirical validations. • Established that Transformers can handle endogeneity in in-context linear regression using instrumental variables, with provable excess loss bounds after pretraining. • Established that a Transformer agent can improve its policy via SARSA or actor-critic in the in-context reinforcement learning setting, with provable convergence guarantees to the optimal parameter manifold.	
Style Learning Induces Semantic Forgetting in Fine-Tuning	Mar 2026 - Present UC Davis
• Investigated why supervised fine-tuning (SFT) can cause more semantic forgetting than reinforcement fine-tuning (RFT), even when teacher demonstrations are semantically correct. • Derived an exact decomposition of semantic and style updates for SFT and RFT in a linear-softmax classification model, identifying teacher-induced style drift as a mechanism for semantic forgetting. • Proved a finite-time lower bound on the forgetting gap between SFT and RFT under population updates, supported by empirical simulations on sequential classification tasks.	
Differential Privacy under Endogeneity	Mar 2025 - Sep 2025 UC Davis
• Proposed a noisy two-stage gradient descent algorithm that addresses endogeneity while guaranteeing ρ-zero-concentrated differential privacy. • Established finite-sample convergence rates and error bounds that quantify the trade-offs among optimization, privacy, and sampling error. • Demonstrated practical privacy-utility trade-offs on both synthetic and real-world datasets.	

Selected Publications

Learning Style, Forgetting Semantics: A Case Study of SFT and RFT on Classification Tasks	2026
<i>H. Liang</i> , Y. Jin, K. Balasubramanian, L. Lai arXiv preprint, arXiv:2610.02437 Paper	
Transformers Provably Implement In-Context Reinforcement Learning with Policy Improvement	2026
<i>H. Liang</i> , L. Lai Advances in Neural Information Processing Systems (NeurIPS 2026) Paper	
Differentially Private Two-Stage Gradient Descent for Instrumental Variable Regression	2025
<i>H. Liang</i> , Y. Jin, K. Balasubramanian, L. Lai International Conference on Learning Representations (ICLR 2026) Paper	
Transformers Handle Endogeneity in In-Context Linear Regression	2024
<i>H. Liang</i> , K. Balasubramanian, L. Lai International Conference on Learning Representations (ICLR 2025) Paper	
ASGARD is A Single-cell Guided Pipeline to Aid Repurposing of Drugs	2023
B. He, Y. Xiao, <i>H. Liang</i> , Q. Huang, Y. Du, Y. Li, D. Garmire, D. Sun, L. Garmire Nature Communications 14, 993 (2023) Paper	

Professional Services

Reviewer: ICLR (2026, 2027), NeurIPS (2026), IEEE Transactions on Information Theory.
Teaching Assistant: EEC 161 (Applied Probability), EEC 018 (Digital System Design), EEC 266 (Information Theory and Coding).
Consultant: CSCAR (Consulting for Statistics, Computing & Analytics Research).
Mentor: Maxwell Lin (Princeton University), Kaleb Kokott (UC Davis).

Honors and Awards

Shu Lin Graduate Student Conference Travel Award	UC Davis, 2026
Summer Graduate Student Researcher Award	UC Davis, 2025
University Graduate Program Fellowship	UC Davis, 2023

Work Experience

BaishanCloud Technology	Jul 2020 - Jul 2021
Data Analyst - Full time	Xiamen, Fujian, China
- Designed optimization models and operational metrics for network bandwidth allocation, minimizing infrastructure cost while satisfying service quality constraints. - Developed Python-based data pipelines integrated with internal APIs to automatically collect, process, and monitor network utilization metrics on a daily basis. - Achieved 14% cost reduction (~\$1.2M annual savings).	

Skills

Programming Languages: Python, R, Matlab, C++, Julia.
Frameworks & Libraries: PyTorch, Hugging Face Transformers, NumPy, SciPy.
Tools: Git, Linux, LaTeX.
Languages: English, Mandarin.